

Clean online latency versus sparsity | Batch size 1

Raw baseline: RGB / grayscale

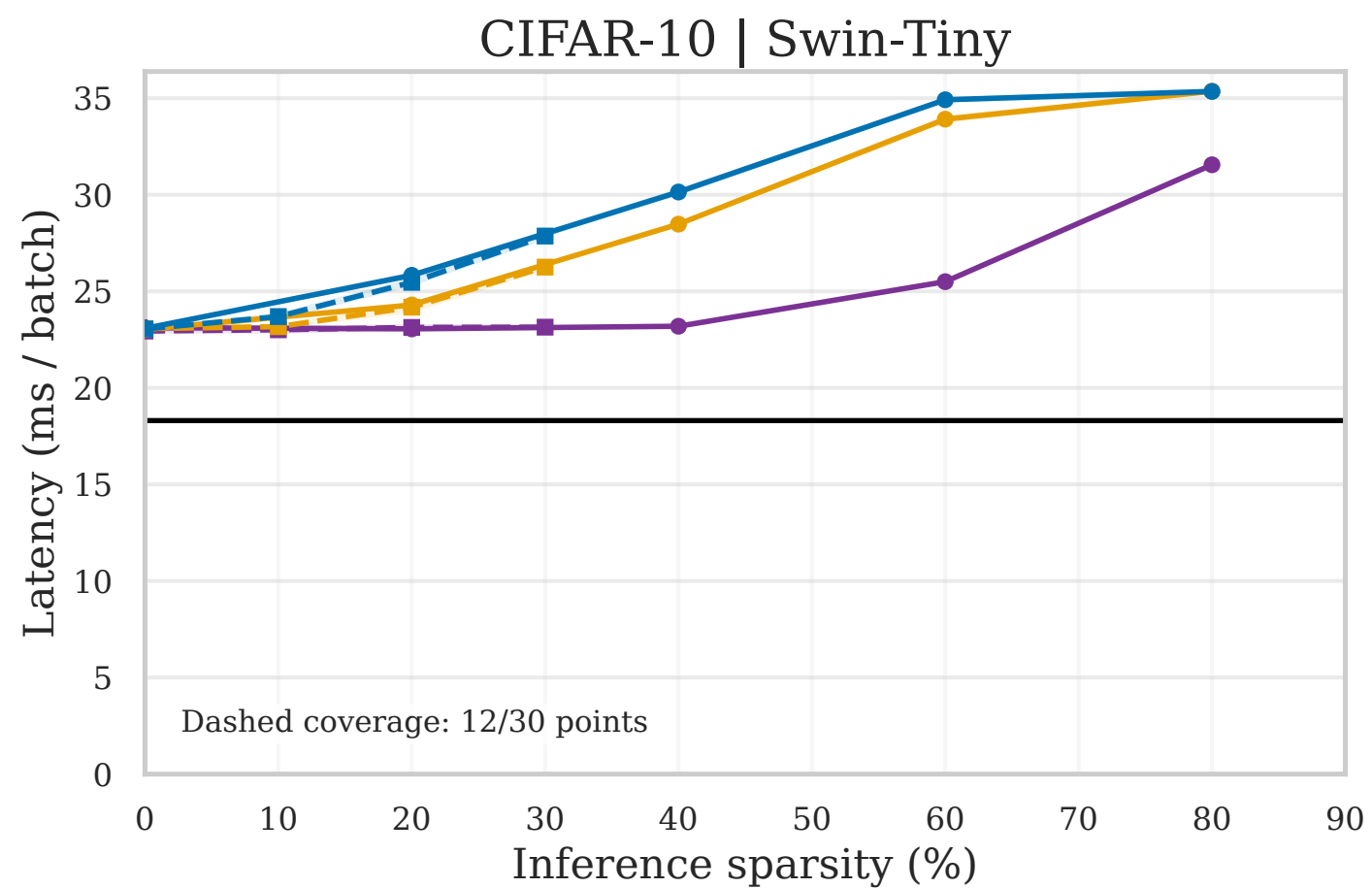
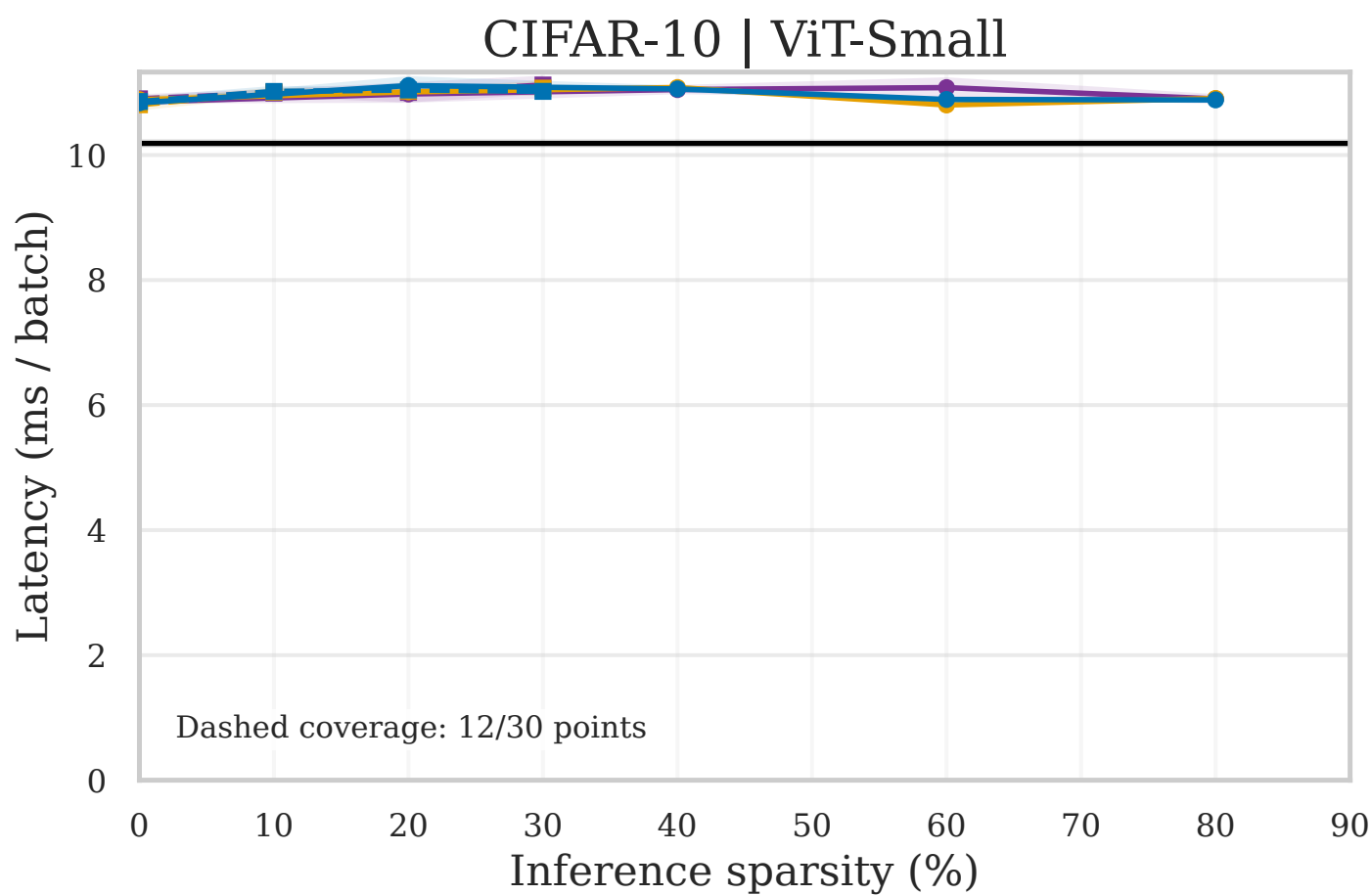
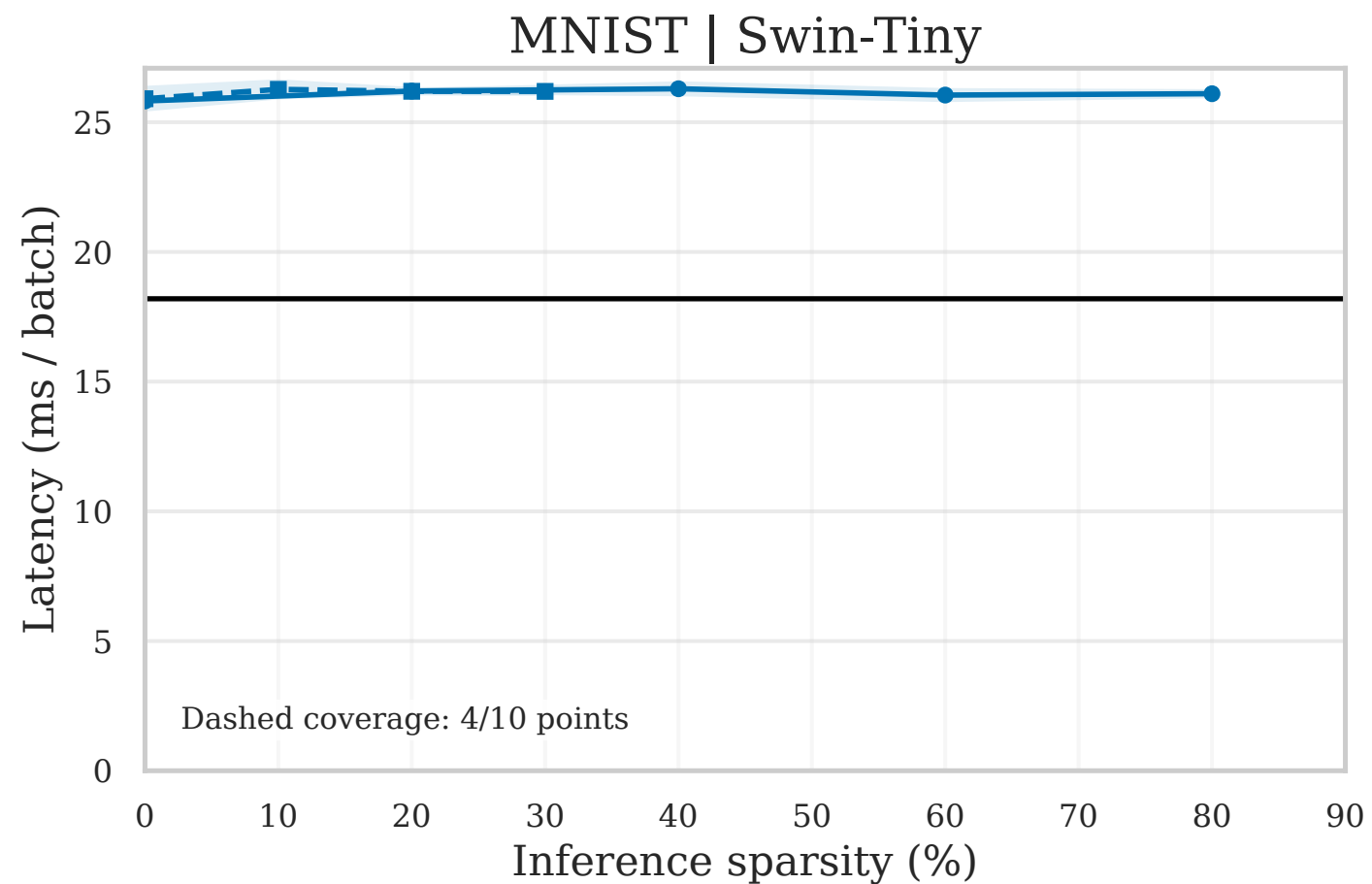
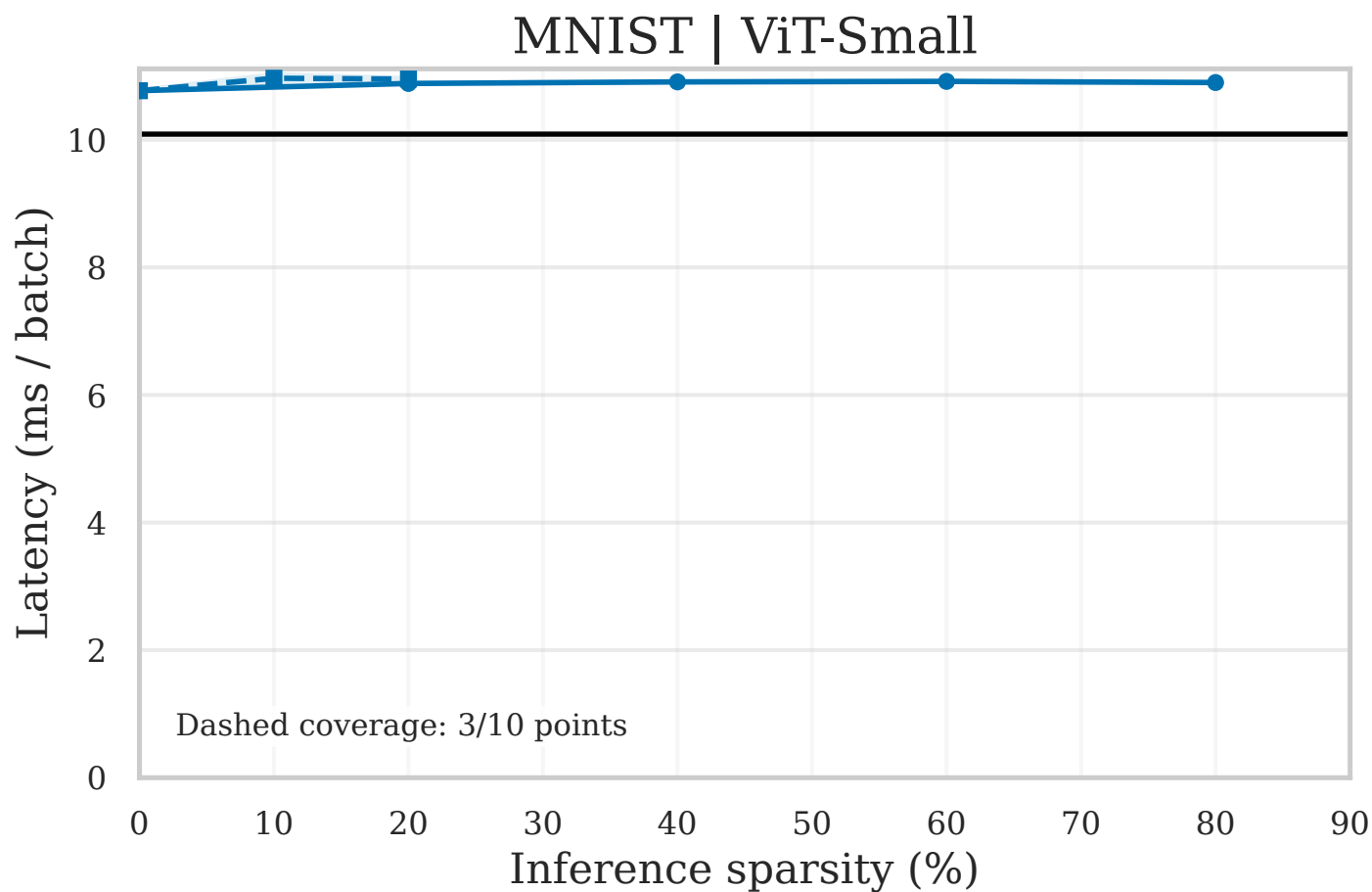
— Color opponency

— Single color

— Grayscale contrast

—●— Solid: trained and tested at matching sparsity

— Dashed: trained dense; tested sparse



Mean of three seed medians; bands show sample SD. Each seed: 50 warm-up + 200 measured batches.

NVIDIA RTX A6000 | AMD EPYC 7413 | FP32 | 4 CPU threads. Missing three-seed points are not connected

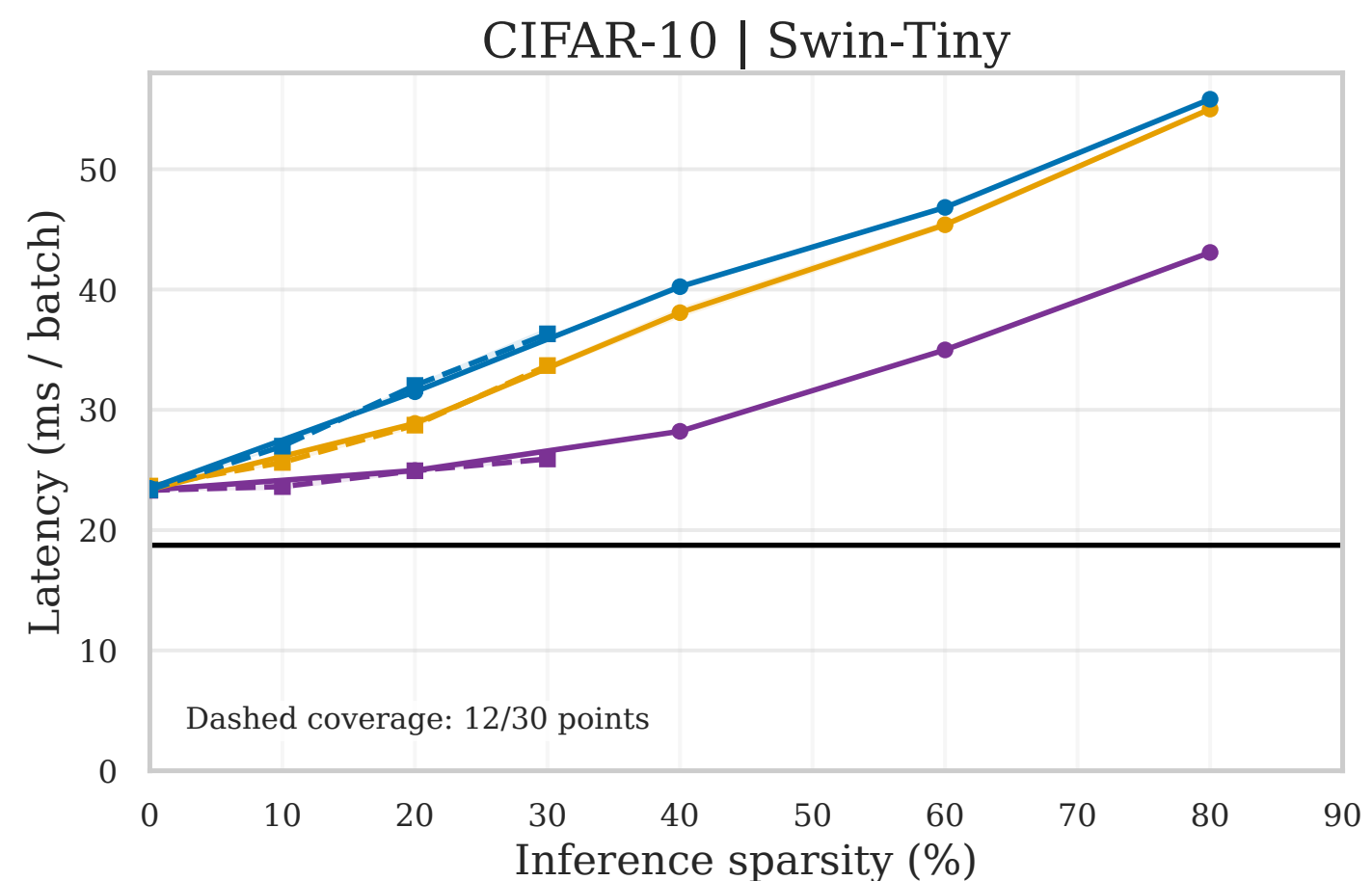
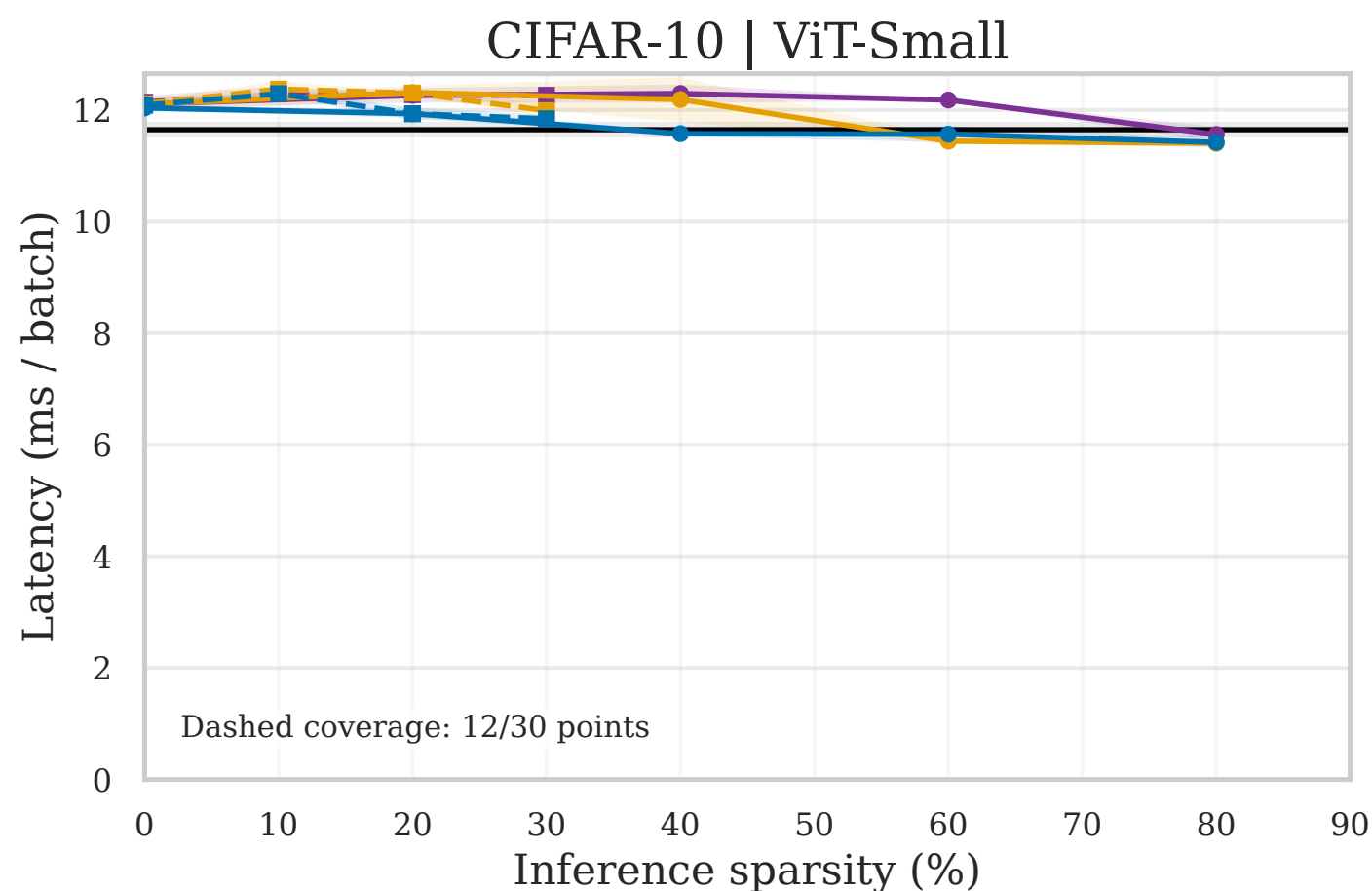
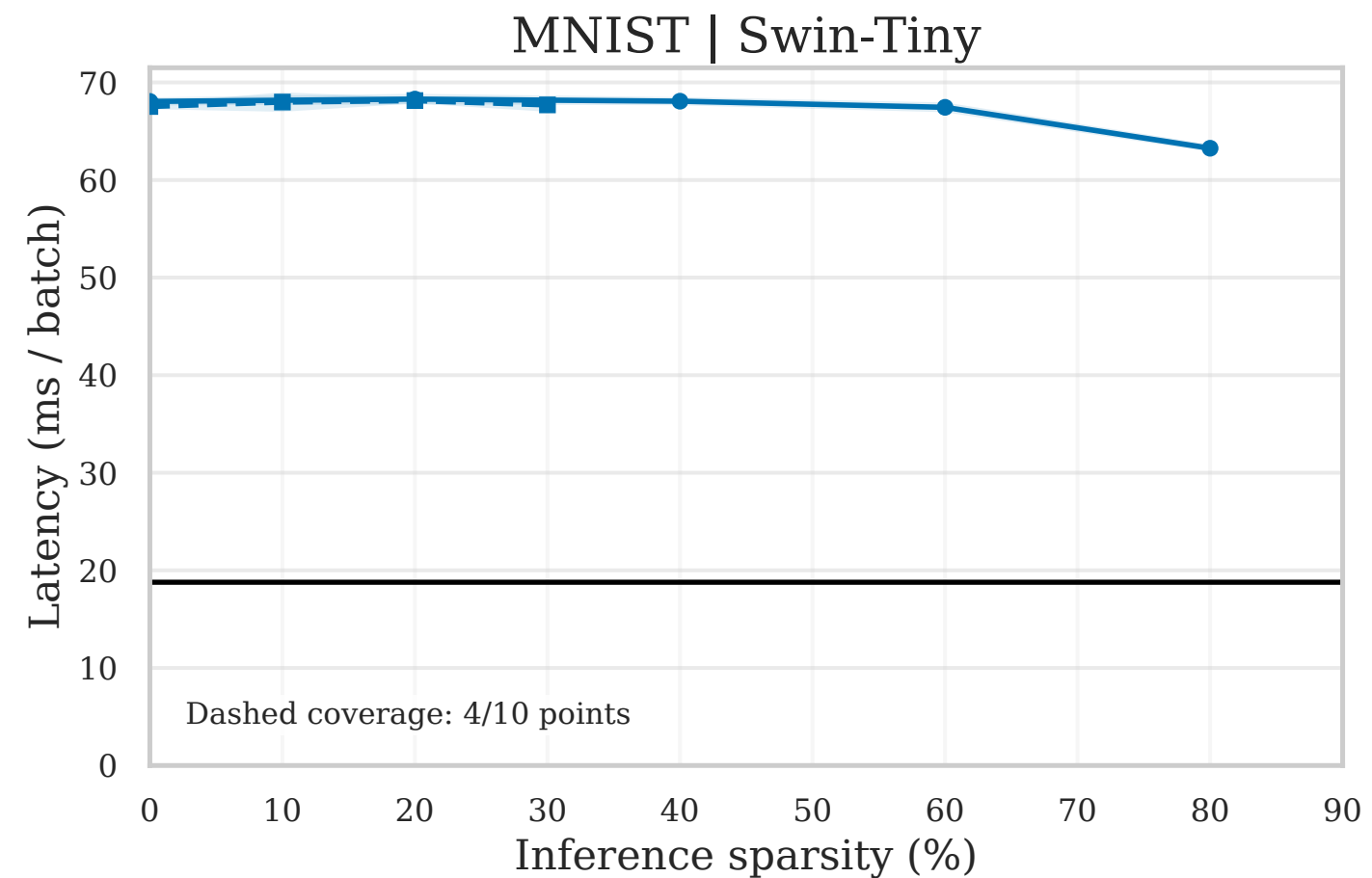
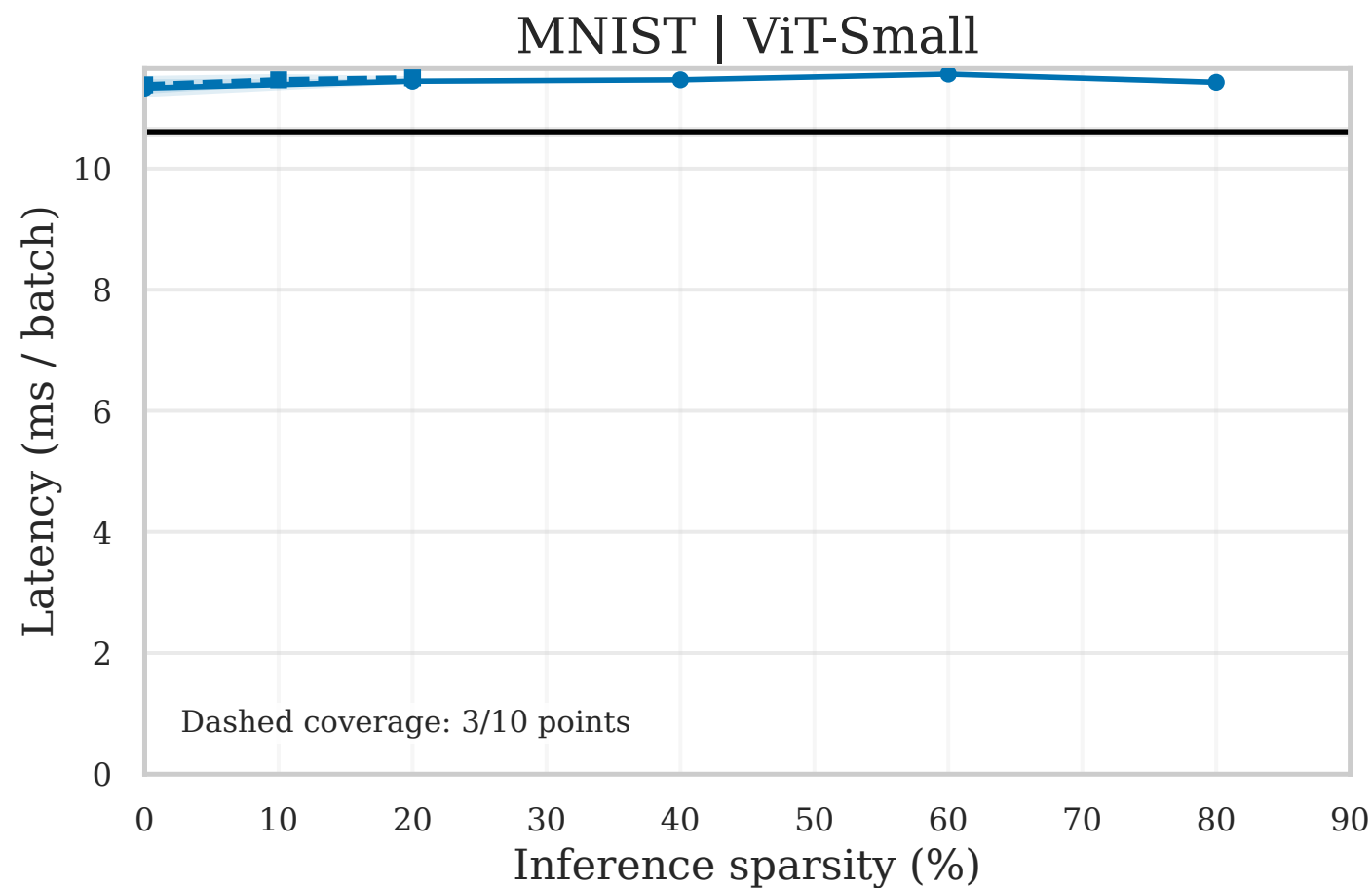
Black line: original dense-row common reference. Exact checkpoints and timing panels match across studies; sessions differ.

Follow-up coverage snapshot: 2026-10-08 11:24 UTC. Raw-pixel sparse sweeps and compact-trained 0% follow-up family are excluded

Clean online latency versus sparsity | Batch size 8

— Raw baseline: RGB / grayscale — Color opponency — Single color — Grayscale contrast

—●— Solid: trained and tested at matching sparsity —■— Dashed: trained dense; tested sparse



Mean of three seed medians; bands show sample SD. Each seed: 50 warm-up + 200 measured batches.

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Black line: original dense-raw common reference. Exact checkpoints and timing panels match across studies; sessions differ.

Follow-up coverage snapshot: 2026-10-08 11:24 UTC. Raw-pixel sparse sweeps and compact-trained 0% follow-up family are excluded.